Spectral Gap and Finite-Level Cusp-Depth Diagnostics of the Weighted Farey Laplacian

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Abstract

Let `F_N` be the Farey fractions of level `N`, and let `L_N` be the unnormalized weighted graph Laplacian with edge weights `w_{uv}=(q_uq_v)^{-2}`. We prove three results. First, the weights admit the exact Ford-circle representation `w_{uv}=exp(-(d(B_infinity,B_u)+d(B_infinity,B_v)))`, where `d` is signed hyperbolic distance between horoballs. The Ford geometry is classical; the weighting identity is the observation used here, and no uniqueness characterization is claimed. Second, a golden-convergent test vector gives `lambda_2(L_N) <= C/N^4` with the explicit admissible constant `C=16 phi^12<5200`. Third, smaller-denominator-parent canonical paths give the reviewed lower bound `lambda_2(L_N) >= c/(N^4 log N)` for an unspecified absolute `c>0`. The selected parent edges form a spanning tree, denominators at least double down its descendant direction, and the resulting congestion is `O(mN^4 log N)`. Thus the gap is bracketed to one logarithmic factor; neither a matching lower bound nor a numerical lower constant is proved.

The BC-010 gap values through `N=128` remain inherited reports. In contrast, the governed reproduction bundle replays the single level `N=80` with Python `3.12.10`, NumPy `2.4.6`, one-thread parity-block diagonalization, generated JSON / CSV tables, and an exact-scope SHA-256 manifest. It gives `m=1967`, lambda_2\approximately\1.95557e-7 in the odd sector and lambda_3\approximately\1.95679e-7 in the even sector, with cross-sector relative separation approximately\6.221e-4. The inherited value `1.2e-3` is not reproduced under the declared 10 percent comparison tolerance and is replaced. The bottom-five projector has mean continued-fraction depth `22.9993` against graph mean `15.9985`, while its mean Ford-horoball depth `2 log q` is `8.75875` against graph mean `7.76982`. These are distinct single-level observables. Reflection supplies invariant parity sectors but no pairing theorem, and no cross-level localization law follows. We close by separating volume and counting-measure isoperimetry. A separate same-system diagnostic at `N=20,40,60,80` is also reported with its prior feasibility probe disclosed; it is descriptive, not preregistered, independent, or an asymptotic theorem.

Scope and evidence boundary

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Continues: doi:10.5281/zenodo.21330770
Concept DOI: doi:10.5281/zenodo.21231880
Mathematical development assistance: Claude (Anthropic), 2026-06-12 `lambda_2(L_N) >= c/(N^4 log N)` for some unspecified absolute `c > 0`. The similarly numbered files `BC-045-claim-signature-20260612.{md,json}` concern an unrelated conditional C-22 bridge and are not evidence for this theorem. External mathematical review remains invited. Nothing in this paper proves or implies the Riemann Hypothesis or full physical SIT. **Machine-readable lower-bound contract:** lambda_$2L_N$ >= c/$N^4 logN$ for some unspecified absolute c > 0. **Evidence classes.** *Proved* means a complete argument appears here or in a specifically located reviewed source. *Replayed measurement* means a value is regenerated by the pinned reproduction bundle with frozen parameters, runtime, commands, outputs, and SHA-256 manifest. *Reported measurement, not replayed* applies only to the inherited BC-010

gap table. *Conjectural* means a stated interpretation or open assertion. The labels are not interchangeable. ## Reader Orientation: The Problem Before the Vocabulary Imagine a finite road network in which every town is reachable, but roads serving increasingly remote towns become extremely weak. The network is connected, yet a disturbance can take a very long time to spread evenly. Two questions arise before any specialized terminology is needed: 1. how slowly can the slowest nonconstant pattern relax as the network grows; and 2. do those slow patterns place unusually large weight on the arithmetically deepest towns? Established spectral graph theory answers the first kind of question with the smallest positive eigenvalue of a graph Laplacian. A test pattern concentrated near a weakly connected vertex gives an upper bound. Conversely, routing every pairwise difference through controlled paths gives a lower bound. Classical Farey and Ford-circle geometry supplies the vertices, adjacency, and a logarithmic depth coordinate. The operation studied here can therefore be stated in plain language: 1. list the reduced fractions whose denominators do not exceed a chosen level; 2. join Farey neighbors; 3. weaken each edge according to the two endpoint denominators; 4. measure the slowest nonconstant relaxation mode; 5. bound that mode from above with one explicit weakly connected witness and from below by controlling the traffic carried by a family of graph paths; 6. inspect finite computed modes using two different arithmetic-depth observables; and 7. keep proved bounds, replayed measurements, unreplayed historical reports, and conjectures in separate evidence classes. Only now do we name the resulting operator the **weighted Farey Laplacian**, the upper witness the **golden-convergent Rayleigh witness**, and the lower argument the **smaller-parent canonical-path construction**. The finite continued-fraction and Ford-depth calculations are **cusp-depth diagnostics**, not a theorem of metric or asymptotic localization. The discriminating tests are equally concrete. A proposed lower theorem fails if a path family cannot supply the stated congestion bound. A numerical localization claim fails if it does not survive pinned replay, residual checks, alternative depth observables, and comparison across levels. A finite table cannot by itself remove the logarithm between the proved upper and lower bounds.

1. Introduction

1.1 Operator and questions

Write

$$F_N=\{p/q \text{ in } [0,1]: \gcd(p,q)=1,\ q<=N\}.$$

Two vertices p/q and r/s are adjacent exactly when

$$|ps-rq|=1, \text{ and } w_{p/q,r/s}=(qs)^{-2}.$$

Let L_N be the corresponding unnormalized Laplacian. The two questions addressed here are geometric and spectral:

1. How can the denominator weight be read in the Ford-horoball geometry?
2. At what rate does the finite spectral gap $\lambda_2(L_N)$ close?

The first answer is an exact identity, not a uniqueness theorem. The second answer is the proved bracket

$$c/(N^4 \log N) <= \lambda_2(L_N) <= C/N^4.$$

with an explicit admissible upper constant and an unspecified absolute positive lower constant. The remaining logarithm is an open problem.

1.2 Main results and evidence boundary

Theorem A (Ford-circle weighting identity; proved). Ford-circle tangency is Farey adjacency, and on every Farey edge the selected weight equals the exponential of minus the total depth of its endpoint horoballs below the standard cusp horoball.

Theorem B (gap upper bound; proved). For every `N>=2`, `lambda_2(L_N)<=16 phi^12 N^{-4}<5200 N^{-4}`.

Theorem C (gap lower bound; internally reviewed-green, AI-assisted, at the stated boundary). For every `N>=2`, `lambda_2(L_N)>=c/(N^4 log N)` for some unspecified absolute `c>0`.

Finite-level evidence. The BC-010 record supplies reported gap values through `N=128`, not replayed in this package. The governed bundle separately replays the `N=80` localization experiment in fixed even and odd reflection sectors and reports both continued-fraction depth and `2 log q`. It resolves the single-level provenance gap but does not prove asymptotic localization or a sharp gap law. A separated same-system four-level diagnostic supplies descriptive comparison across `N=20,40,60,80` without promoting a trend.

1.3 Source locations and related work

Ford defines the circle at `p/q` with radius `1/(2q^2)` in Section 1, pp. 587-588, and proves the tangency determinant calculation in Theorem 1 on p. 588; his Sections 2-4, pp. 588-591, connect adjacency and Farey series. Springborn gives the modern horocycle normalization in Sections 5-7 and the signed-distance formula in Proposition 6.2, equation (24), pp. 12-15 of arXiv:1702.05061. Athreya, Chaubey, Malik, and Zaharescu study related Farey-Ford polygon geometry.

The path-length congestion inequality used below is the uniform-measure, continuous-time Laplacian adaptation of Diaconis and Stroock (1991), Proposition 1', p. 38. Sinclair (1992) is the multicommodity-flow reference. The normalized Cheeger material is located in Chung (1997), Chapter 2, especially Sections 2.2-2.3. Series (1985) supplies modular-surface and continued-fraction context; Hurwitz (1891) supplies the approximation extremality behind the golden-convergent witness.

Within the deposited series, Paper 1 (DOI 10.5281/zenodo.21231873) defines the operator context, Paper 3 (DOI 10.5281/zenodo.21231877) proves that its RH-equivalent deficit is spectrum-blind, Paper 7 (DOI 10.5281/zenodo.21231887) studies a weight family, Paper 8 (DOI 10.5281/zenodo.21231891) is the computational companion, and Paper 9 (DOI 10.5281/zenodo.21231894) is the survey propagation target. The replay used in Section 6 lives only in this correction package; no companion record is treated as substitute evidence.

The new claims are bounded as follows: the Ford geometry itself is classical; the exact identification of this chosen edge weight with total endpoint depth is the geometric observation; the two-sided gap bracket is theorem-grade at the constants stated here; and all localization language beyond finite replayed statistics is conjectural. A broader prior-art scan for weighted Farey isoperimetry remains a future novelty-review task; this preprint makes no uniqueness claim.

1.4 Roadmap

Section 2 defines the operator and proves gap positivity. Section 3 proves the Ford-circle identity. Section 4 proves the upper bound. Section 5 gives a self-contained canonical-path inequality and the corrected lower bound. Section 6 specifies the pinned `N=80` replay and the separated four-level diagnostic while retaining their symmetry and inference boundaries. Section 7 separates two isoperimetric normalizations. Section 8 lists open problems, Section 9 states limitations, and Appendix A records the cross-paper correction contract.

2. The finite weighted Farey Laplacian

2.1 Definitions

Let

$$m=m(N)=|F_N|=1+\sum_{q\le N} \varphi(q).\quad \textbf{[1]}$$

Thus $m=(3/\pi^2)N^2+O(N \log N)$. For a vertex v, write $\mathrm{den}(v)$ for its reduced denominator. Set

$$w_{uv}=(\mathrm{den}(u)\mathrm{den}(v))^{-2} \text{ for } u\sim v.\quad \textbf{[2]}$$

and $W(u)=\sum_{v\sim u}w_{uv}$. The Laplacian is

$$(L_N f)(u)=\sum_{v\sim u}w_{uv}(f(u)-f(v)).\quad \textbf{[3]}$$

It is real symmetric and positive semidefinite, with

$$\langle f,L_Nf\rangle=\sum_{\text{unordered } \{u,v\}}w_{uv}(f(u)-f(v))^2.\quad \textbf{[4]}$$

Write

$$0=\lambda_1\le\lambda_2\le\ldots\le\lambda_m.\quad \textbf{[5]}$$

2.2 Connectivity and the kernel

Proposition 2.1 (connectivity and gap positivity; proved). The level-N Farey graph is connected for every $N\ge1$. Consequently $\ker L_N=\mathrm{span}\{1\}$, and $\lambda_2(L_N)>0$ for $N\ge2$.

Proof. Every nonendpoint p/q with $q\ge2$ is the mediant of two Farey parents whose denominators are positive, less than q, and sum to q. Following either parent strictly decreases the denominator and eventually reaches $0/1$ or $1/1$; those endpoints are adjacent. Positive edge weights then imply that zero Dirichlet energy is equivalent to constancy on the connected graph. ■

The infinite-volume statements in Paper 1 are contextual and are not inputs to Theorems A-C.

3. Ford circles and the weighting identity

Let $B_{p/q}$ be the Ford horoball of Euclidean diameter q^{-2} tangent to the real axis at p/q, and let $B_\infty=\{z: \mathrm{Im}\, z\ge1\}$. Signed distance is positive for disjoint horoballs, zero at tangency, and negative on overlap.

Theorem 3.1 (Ford-circle weighting identity; Theorem A; proved). For reduced, distinct rational points:

1. $d(B_\infty,B_{p/q})=2 \log q$;
2. $d(B_{p/q},B_{r/s})=2 \log|ps-rq|$;
3. Ford circles are tangent iff $|ps-rq|=1$, equivalently iff their fractions are Farey adjacent;
4. on each Farey edge $\{u,v\}$, $w_{uv}=\exp(-(d(B_\infty,B_u)+d(B_\infty,B_v)))$. **[6]**

Proof. The top of $B_{p/q}$ has Euclidean height q^{-2}. The vertical hyperbolic distance from height 1 to height q^{-2} is $\int_{q^{-2}}^1 dy/y=2 \log q$. Springborn's Proposition 6.2, equation (24), gives the second formula in the same normalization. Since the determinant is a nonzero integer for distinct reduced fractions, it has absolute value at least one and vanishes from the distance exactly at tangency. Ford's Theorem 1 gives the equivalent Euclidean circle calculation. The last assertion is the identity

$$\exp(-(2 \log q_u+2 \log q_v))=(q_uq_v)^{-2}.\ \blacksquare$$

Remark 3.2 (claim boundary). Parts 1-3 are classical. Part 4 identifies the selected weights with one natural exponential function of endpoint depth. It does not prove that no other depth-dependent weighting is natural, nor does it characterize the weight uniquely from invariance axioms.

4. Gap upper bound

Let `phi=(1+sqrt(5))/2`, and let `F_k` denote Fibonacci numbers with `F_1=F_2=1`.

Lemma 4.1 (neighbors of a golden convergent; proved). Choose `k>=2` so that `F_{k+1}<=N<F_{k+2}`, and set

`v*=F_k/F_{k+1}`. **[7]**

Every Farey neighbor `r/s` of `v*` in `F_N` satisfies `s>=F_{k-1}>N/phi^4`, and `v*` has at most eight such neighbors.

Proof. Adjacency gives

`F_ks-rF_{k+1}=+1 or -1`, so `s=+F_k^{-1} or -F_k^{-1} (mod F_{k+1})`. **[8]**

Cassini's identity `F_{k+1}F_{k-1}-F_k^2=(-1)^k` identifies the two positive residue classes as `F_k` and `F_{k-1}`. Their least positive representatives are therefore at least `F_{k-1}`. Binet's formula gives `F_{k-1}/F_{k+2}>=phi^{-4}`, hence `s>N/phi^4`.

Because `N<F_{k+2}<2F_{k+1}`, each residue class has at most two representatives below `N`. Allowing both determinant signs gives the safe upper count `8`. This count is deliberately conservative; only boundedness is used. ■

Theorem 4.2 (upper bound; Theorem B; proved). For every `N>=2`,

`lambda_2(L_N) <= 8 phi^12 N^{-4}(1-1/m)^{-1}    <= 16 phi^12 N^{-4}<5200 N^{-4}`. **[9]**

Thus `C=16 phi^12` is an explicit uniform admissible upper constant.

Proof. Let `psi=delta_{v*}-m^{-1}1`, so `psi` is orthogonal to the constant vector, `||psi||^2=1-1/m`, and `<psi,L_Npsi>=W(v*)`. Lemma 4.1 gives at most eight neighbors with denominator greater than `N/phi^4`. Binet's formula also gives `den(v*)=F_{k+1}>N/phi^2`. Hence

`W(v*)<=8(N/phi^2)^{-2}(N/phi^4)^{-2} =8 phi^12 N^{-4}`. **[10]**

The Rayleigh principle gives the first inequality. Since `m>=2`, `(1-1/m)^{-1}<=2`, giving the uniform constant. ■

Remark 4.3. Golden convergents make all low-level neighbor denominators large and therefore furnish a natural extremal test vector. The proof does not claim that this witness is unique.

4.1 Reported finite-level gaps

The following table reproduces BC-010's reported output. The source file bc010_gap_and_mass_profile.py now exists beside BC-010-gap-law-and-mass-profile-20260612.md, but this correction package does not pin its Python / NumPy environment, command transcript, outputs, or hashes and has not replayed the dense eigensolves.

`N`	`m`	reported `lambda_2`	successive exponent
8	23	`2.305e-03`	`-`
16	81	`1.261e-04`	`-4.192`
32	325	`7.690e-06`	`-4.036`
64	1261	`4.778e-07`	`-4.009`
128	5023	`2.982e-08`	`-4.002`

These reports are compatible with both `N^{-4}` and `N^{-4}/log N` over the displayed range. They do not prove the sharp asymptotic.

Conjecture 4.4 (sharpness; conjectural). `lambda_2(L_N)` is comparable to `N^{-4}`. This is strictly stronger than Theorem 5.3 and remains open.

5. Gap lower bound

5.1 Canonical-path inequality

Proposition 5.1 (path congestion; proved). Let `G` be a connected weighted graph on `m` vertices with unnormalized weighted Laplacian `L`. For each ordered pair `x!=y`, choose a simple path `gamma_xy`. Then

`lambda_2(L) >= 2m / rho`. **[11]**

where

`rho=max_e w_e^{-1} sum_{(x,y): e in gamma_xy}|gamma_xy|`. **[12]**

This is the uniform-counting-measure adaptation of Diaconis-Stroock (1991), Proposition 1', p. 38.

Proof. For `f` orthogonal to `1`,

`sum_{x!=y}(f(x)-f(y))^2=2m||f||^2`. **[13]**

Telescoping on `gamma_xy` and applying unweighted Cauchy-Schwarz gives

`(f(x)-f(y))^2 <=|gamma_xy| sum_{e in gamma_xy}(df_e)^2 =sum_{e in gamma_xy}|gamma_xy|w_e^{-1}[w_e(df_e)^2]`. **[14]**

After summing over ordered pairs and exchanging finite sums,

`2m||f||^2<=rho sum_e w_e(df_e)^2=rho<f,Lf>`. **[15]**

Minimization over nonzero `f` orthogonal to `1` proves the claim. ■

5.2 Audit history

The retracted route selected the larger-denominator parent. In the small-complementary-parent case, especially denominator `1`, its subtree interval was only of order `1/Q`; the displayed proof supported the older `N^{-6}` floor, not the advertised `N^{-5}` refinement. BC-013 recorded that downgrade.

The binding repair is BC-045-c24-audit-and-sharp-gap-lower-bound-20260612.md. Its Section 2 replaces the route by the smaller-denominator parent; its Section 4 and second-clearance note mark the repaired rate internally reviewed-green (AI-assisted). The clearance also states that the lower constant remains unspecified. Bare references to `BC-045` are not adequate because the separate BC-045-claim-signature-20260612.* files sign an unrelated C-22 bridge.

5.3 Corrected lower bound

Theorem 5.2 (corrected lower bound; Theorem C; internally reviewed-green, AI-assisted, at this exact boundary). There is an unspecified absolute constant $c>0$ such that, for every $N>=2$,

$$\lambda_2(L_N)>=c/(N^4 \log N). \quad \textbf{[16]}$$

Here $\log$ is natural logarithm. A change of base is absorbed into the unspecified constant.

Proof. **Smaller-parent tree.** Every nonroot $v=p/Q$ has two Farey mediant parents whose denominators sum to Q. Let $sp(v)$ be the parent with smaller denominator s, so $s<=Q/2$. Equal parent denominators occur only for $v=1/2$; choose $sp(1/2)=0/1$.

There are $m-2$ selected edges $\{v,sp(v)\}$, one for each nonroot vertex. Add the root connector $\{0/1,1/1\}$. Every selected nonroot edge is a Farey edge, and repeated parent selection strictly decreases the denominator until one of the roots is reached. Thus the selected graph is connected. It has $m-1$ edges and no cycle: any nonroot selected-edge cycle would contradict strict denominator descent, and the single root connector cannot create a cycle without a second route between the roots. The selected graph is therefore a spanning tree.

For each ordered pair, let γ_{xy} be the unique simple tree path. If the two parent chains meet before a root, the path stops at their first common ancestor and does not repeat their common tail.

Depth. Every parent step at least halves the denominator. Consequently each rootward chain has at most $\log_2 N$ edges and

$$|\gamma_{xy}|<=2 \log_2 N+1. \quad \textbf{[17]}$$

Descendant component. For a nonroot v of denominator Q, let $T(v)$ be the component below v after deleting $\{v,sp(v)\}$; equivalently, it is the set of vertices whose selected-parent chain contains v. If $sp(u)=v$, the other parent of u has denominator at least Q; otherwise it would have been selected. Hence $den(u)>=2Q$.

For a descendant chain

$$v=v_0,v_1,...,v_k=u, \text{ with } sp(v_i)=v_{i-1} \text{ and } Q_i>=2^iQ. \quad \textbf{[18]}$$

write $Q_i=den(v_i)$. Inductively, $Q_i>=2^iQ$. Consecutive terms are Farey adjacent, so

$$|v_i-v_{i-1}|=1/(Q_iQ_{i-1})<=2/(4^iQ^2). \quad \textbf{[19]}$$

Therefore

$$|u-v|<=\sum_{i>=1}2/(4^iQ^2)=2/(3Q^2)<1/Q^2. \quad \textbf{[20]}$$

All of $T(v)$ lies in one interval of length less than $2/Q^2$. We retain the looser length $4/Q^2$ to match the reviewed packet. Consecutive elements of F_N are separated by at least $1/N^2$, hence

$$|T(v)|<=1+4N^2/Q^2. \quad \textbf{[21]}$$

This is a telescoping distance argument; it makes no assumption about the number of descendants at a fixed depth.

Congestion. Fix the tree edge $e=\{v,sp(v)\}$. A simple tree path uses e iff exactly one endpoint is in $T(v)$. Thus the exact number of ordered pairs crossing e is $2|T(v)|(m-|T(v)|)$, at most $2m|T(v)|$. Thus

$$\text{number of ordered pairs crossing } e = 2|T(v)|(m-|T(v)|). \quad \textbf{[22]}$$

Since $w_e^{-1}=(Qs)^2$,

$$w_e^{-1} \sum_{\gamma_{xy} \text{ containing } e}|\gamma_{xy}| <=(Qs)^2\, 2m(1+4N^2/Q^2)(2 \log_2 N+1) =2m(2 \log_2 N+1)(Q^2s^2+4s^2N^2). \quad \textbf{[23]}$$

Because $s<=Q/2<=N/2$, both terms in the last parenthesis are bounded by an absolute multiple of N^4. Also $2 \log_2 N+1$ is bounded by an absolute multiple of $\log N$ for $N>=2$. Every nonroot selected edge therefore has congestion at most $C_0mN^4 \log N$ for an absolute C_0.

The root connector has inverse weight 1, carries at most m^2 ordered pairs, and has congestion $O(m^2 \log N)=O(mN^2 \log N)$ because $m=O(N^2)$. It is subdominant. Enlarging an absolute congestion constant to cover the finite initial levels, Proposition 5.1 gives

$$\lambda_2(L_N)>=2m/(C_1mN^4 \log N)=c/(N^4 \log N). \quad [24]$$

for some absolute $c>0$. This argument establishes existence only; it does not certify a numerical value. ■

Remark 5.3 (asymptotic boundary). The lower and upper bounds differ by a factor $\log N$. The surviving logarithm is the maximum path-length factor in this canonical-path estimate. The argument does not decide whether another routing, a flow, or an isoperimetric method can remove it.

6. Pinned finite-level localization replay

This section separates a proved symmetry statement from a deterministic single-level experiment. The replay closes the provenance gap for $N=80$; it does not prove cross-level or asymptotic localization.

Definition 6.1 (two depth observables). For a regular continued fraction $p/q=[0;a_1,...,a_k]$, set

$$\mathrm{depth\_CF}(p/q)=\sum_i a_i. \quad [25]$$

The two finite expansions of a rational have the same partial-quotient sum, so this statistic is well-defined. It records Stern-Brocot path depth. The geometric observable supplied by Theorem 3.1 is instead

$$\mathrm{depth\_H}(p/q)=2\log q. \quad [26]$$

The replay computes both vertex observables. No monotone functional relation between them is assumed, and a statistic for one is not relabeled as a statistic for the other.

Lemma 6.2 (reflection sectors; proved). The map σ acts by

$$\sigma(p/q)=(q-p)/q, \text{ with } \sigma L_N=L_N\sigma. \quad [27]$$

It is a weight-preserving automorphism of the level-N Farey graph, and the space is the orthogonal sum of even and odd invariant subspaces.

Proof. Reflection preserves denominators, and $|(q-p)s-(s-r)q|=|rq-ps|$, so it preserves adjacency and edge weights. The Laplacian therefore commutes with the permutation matrix of σ. Since $\sigma^2=I$ and σ is self-adjoint, its $+1$ and -1 eigenspaces are orthogonal and invariant under L_N. ■

Warning 6.3 (no forced pairing). Lemma 6.2 does not imply equal or nearby eigenvalues across the two sectors. At $N=2$, direct diagonalization gives spectrum $\{0,3/4,9/4\}$; the nonzero even eigenvalue is $3/4$ and the odd eigenvalue is $9/4$. Any small cross-sector splitting at $N=80$ is numerical structure, not a symmetry consequence.

6.1 Frozen experiment contract

The bundle reproducibility/localization-n80/ fixes the following choices:

1. $N=80$, with the 1967 reduced fractions in $[0,1]$ ordered as the Farey sequence and independently checked against $1+\sum_{q<=80}\varphi(q)$;
2. adjacency $|ps-rq|=1$, weights $(qs)^{-2}$, and the unnormalized float64 Laplacian from Section 2;

3. orthonormal reflection-orbit bases, giving an even block of dimension `984` and an odd block of dimension `983`;
4. `numpy.linalg.eigh` under Python `3.12.10`, NumPy `2.4.6`, and forced one-thread BLAS-related environment variables;
5. the bottom projector as the average of squared amplitudes for global zero-based modes `1,2,3,4,5`, and the middle control as five modes beginning at `floor(m/2)=983`;
6. sign-invariant squared-amplitude observables, corrected closed endpoint position bins, and exact integrated Gauss-Kuzmin bin masses;
7. generated `summary.json`, complete mode and vertex tables, separate CF-depth and denominator-depth distributions, runtime identity, and an exact-scope SHA-256 manifest.

There is no fitted parameter and no random seed. The replay script refuses a Python or NumPy version mismatch. Its maximum block eigenpair residuals are `8.55e-16` (even) and `2.17e-15` (odd), while the reflection commutator error is `6.66e-16`. These diagnostics are numerical consistency checks, not proof of an asymptotic claim.

6.2 Parity-resolved spectrum

Replayed data 6.4 (`N=80`; evidence class). The two lowest nonzero eigenvalues are

quantity	replayed value	parity
`lambda_2`	`1.95557e-7`	odd
`lambda_3`	`1.95679e-7`	even
lambda_3/lambda_2-1	`6.221e-4`	cross-sector relative separation

The reported cross-sector statistic is

delta_23=lambda_3/lambda_2-1. **[28]**

The five lowest nonzero global modes alternate odd, even, odd, even, odd. The residual-based interval for the cross-sector relative separation is approximately `[6.2211005e-4, 6.2214103e-4]`; full float64 values remain in the raw replay output. The inherited manuscript's `1.2e-3` relative split is not reproduced under the declared 10 percent relative comparison tolerance and is withdrawn. The BC-013 packet preserves only rounded outputs, and the named original generator is not archived in the package or canonical source tree. This replay is reconstructed from that packet and its rounded outputs; the middle-mode selection and legacy-bin implementation are explicit Draft 5 choices. The corrected value above is a finite-level cross-sector statistic, not a pairing law or tunneling estimate.

6.3 CF depth and Ford-horoball depth

For a normalized vector `psi`, write

`E_psi[d]=sum_v |psi(v)|^2 d(v).` **[29]**

For a five-mode subspace, the table uses the average of these probability densities, equivalently the diagonal of the normalized rank-five spectral projector. This makes the group statistic invariant under an orthogonal change of basis within that selected subspace.

Replayed data 6.5 (`N=80`; evidence class).

mass distribution	mean `depth_CF`	mean depth_H=2\log\q
uniform graph mass	15.9984748348	7.7698188216
Fiedler mass	22.9999962839	8.7640519274
bottom-five projector mass	22.9993438135	8.7587515059
middle-five projector mass	11.8217630489	7.8655961708

The rounded CF-depth values reproduce the inherited `23.0`, `11.8`, and `16.0`. The direct geometric statistic is new to this correction package: the bottom-five mean `2 log q` exceeds the graph mean by about `0.989`, while the middle control is close to the graph mean. This is evidence of a finite-level shift toward larger denominators under the selected projector. It is not, by itself, a concentration theorem: a mean does not specify tails, support size, or behavior as `N` grows. The complete per-vertex and per-denominator masses are retained in the generated CSVs so later work can define and test those stronger properties without reconstructing the raw matrix convention.

6.4 Position-profile negative control

The corrected 20-bin Fiedler profile includes both endpoints and compares each bin to the exact integral of `1/((1+x)log 2)`. Its Pearson correlation with the Gauss-Kuzmin bin masses is `-0.06264257`; the Fiedler mass below `x=0.25` is `2.09e-16`, while the reference density assigns that interval mass log_2(1.25)=0.32192809. Reapplying the original midpoint and endpoint semantics gives correlation `-0.06262494`, reproducing the inherited `-0.06` at its displayed precision.

This single binned correlation rejects only the naive identification used in that finite experiment. It does not establish global non-equidistribution, identify a transfer operator, or support an RH implication.

6.5 Data and figure disposition

The replay emits plot-ready tables rather than a manuscript image:

- `outputs/modes.csv` records every eigenvalue, parity, and depth mean;
- outputs/vertices.csv records every rational and selected projector mass;
- outputs/cf-depth-distribution.csv and outputs/horoball-depth-by-denominator.csv separate the two depth axes;
- outputs/position-profile.csv records corrected and legacy bin semantics.

No figure is required to evaluate the numerical statements above, so the earlier unresolved figure placeholder is explicitly omitted at Draft 5. Any later rendering must be generated from these tables and captioned as a pinned, single-level, evidence-class experiment.

6.6 Separated same-system multilevel diagnostic

The same reflection-block implementation was also run at four levels under the pinned Python `3.12.10`, NumPy `2.4.6`, and one-thread contract. A timing and feasibility probe used the same four levels before the protocol was frozen. The probe is disclosed; this is neither independent reproduction nor a preregistered confirmation.

`N`	vertices	`lambda_2`	sector	`lambda_3`	relative separation	adjacent descriptive exponent	`N^4 lambda_2`
20	129	5.103362677e-5	odd	5.159935032e-5	1.108530953e-2	-	8.165380284
40	491	3.140725513e-6	odd	3.149258783e-6	2.716974132e-3	4.022278405	8.040257314
60	1103	6.186591879e-7	odd	6.194122358e-7	1.217225701e-3	4.006891194	8.017823075
80	1967	1.955569597e-7	odd	1.956786206e-7	6.221255387e-4	4.003387613	8.010013068

Every level passes the Farey vertex-count identity, reflection-involution, Laplacian row-sum, positive-mode, eigenpair-residual, and orthonormality guards. Across these four levels, `N^4 lambda_2` changes from about `8.165` to `8.010`, and the displayed adjacent finite-difference exponents lie near four. Those are descriptive finite-level observations. Because the levels share one implementation and environment and were used in the disclosed feasibility probe, the table does not establish a sharp `N^-4` asymptotic, remove the logarithm from the proved lower bound, prove parity pairing, or establish asymptotic localization.

The historical BC-010 table and this governed multilevel table remain separate. They have different levels, provenance, replay status, and evidentiary roles.

Remark 6.7 (cusp interpretation; conjectural beyond the tables). The governed replay reports both the `2 log q` distribution and plot-ready denominator groups. It therefore supports the bounded statement that selected bottom-mode mass lies at greater mean Ford-horoball depth than uniform graph mass at `N=80`. Metric cusp localization would require a declared tail or support-concentration criterion and replication over multiple levels. No such cross-level theorem is claimed here, and CF depth is not substituted for Ford-horoball depth. The separated four-level diagnostic motivates that question without answering it.

The lower gap bound independently implies the operator-norm estimate

`||L_N^+||=1/lambda_2(L_N)=O(N^4 log N)` on the nonconstant subspace. **[30]**

This controls a worst-case inverse norm, not the arithmetic deficit in Paper 3, which is spectrum-blind. The replay supplies no RH conclusion.

7. Two isoperimetric normalizations

For `S` a proper nonempty subset of `F_N`, let

`w(partial S)=sum_{u in S,v notin S,u~v}w_{uv}`. **[31]**

7.1 Volume conductance

Set `vol(S)=sum_{v in S}W(v)` and

h_vol(N)=min\w(partial\S)/vol(S). **[32]**

where the minimum is over `0<vol(S)<=vol(F_N)/2`.

Let `L_norm=D^{-1/2}L_ND^{-1/2}` and denote its first nonzero eigenvalue by `lambda_2_norm`.

Proposition 7.1 (normalized Cheeger inequality; proved from the standard weighted theorem).

`lambda_2_norm/2<=h_vol(N)<=sqrt(2 lambda_2_norm)`. **[33]**

This is Chung (1997), Chapter 2, Sections 2.2-2.3, in the present weighted notation.

The normalized and unnormalized eigenvalues satisfy

`lambda_2(L_N)/max_vW(v)<=lambda_2_norm <=lambda_2(L_N)/min_vW(v)`. **[34]**

Justification. Apply the Courant-Fischer characterization to the generalized pair `(L_N,D)`. Since `min_vW(v)||f||^2<=<f,Df><=max_vW(v)||f||^2` for every `f`, the two Rayleigh quotients compare on every subspace, including the quotient by the common one-dimensional kernel. ■

Because the degrees vary by powers of `N`, these comparisons lose powers and do not turn Theorems 4.2 and 5.2 into a sharp estimate of `h_vol`.

7.2 Counting-measure expansion

The unnormalized Rayleigh quotient uses counting measure. Its corresponding cut quantity is

`h_count(N)=min w(partial S)/min(|S|,|F_N\S|).` **[35]**

For the singleton `S={v*}` used in Theorem 4.2, `w(partial S)/|S|=W(v*)=O(N^{-4}).` In contrast, its volume conductance is w(partial\S)/vol(S)=1. The upper-bound witness is therefore an isoperimetric observation for `h_count`, not for `h_vol`.

Tree-descendant sets `T(v)` are useful candidate cuts for either normalization, but their boundaries contain non-tree Farey edges. A minimum over selected-parent tree edges is not automatically a minimum for the full Farey graph.

Question 7.2 (two-normalization extremal cuts). Determine the orders of `h_count(N)` and `h_vol(N)`, identify their extremal cuts, and decide whether golden-convergent singleton or descendant sets are extremal for either quantity. No golden-subtree extremality theorem or conjectural status is asserted without a reproducible cut census and a completed literature scan.

8. Open problems

Open Problem 8.1 (remove or justify the logarithm). Decide whether there is an absolute `c'>0` with `lambda_2(L_N)>=c'N^{-4}` for all `N`, or whether `lambda_2(L_N)N^4` has zero lower limit. Candidate methods include fractional multicommodity flow and isoperimetry. The finite reported exponents do not distinguish the alternatives.

Open Problem 8.2 (isoperimetry). Determine both cut quantities in Section 7 with their normalizations held fixed. Establish any implication for the unnormalized spectral gap without hiding degree-spread losses.

Open Problem 8.3 (independent, preregistered multilevel localization). The governed bundle pins the graph constructor, matrix convention, eigensolver, parity labeling, sign-invariant observables, binning, eigenpairs, both depth statistics, environment, commands, outputs, and hashes at `N=80`. It also contains the disclosed same-system diagnostic at `N=20,40,60,80`. The next step is an independent clean-environment reproduction with levels and trend statistics declared before computation. Only then should an asymptotic localization theorem be posed.

Open Problem 8.4 (metric cusp localization). Determine whether the Fiedler or bottom-projector mass concentrates in `2 log q`, or an equivalent horoball-depth observable, on a vanishing fraction of vertices. The governed single-level mean shift and the separated multilevel gap diagnostic are not this theorem.

9. Limitations and release boundary

1. The lower theorem authorizes an unspecified absolute constant only. It does not authorize a numerical value or a pure `N^{-4}` lower bound.
2. The upper and lower estimates differ by `log N`; the sharp asymptotic is open.
3. The BC-010 gap table is preserved as an inherited historical report and is not replay-pinned in this package. It is not used as theorem evidence. The `N=80` localization experiment is pinned, and the separate same-system `N=20,40,60,80` diagnostic is reproducible under the declared environment.
4. Reflection symmetry gives parity sectors but no forced cross-sector pairing; the inherited `1.2e-3` split was not reproduced and is replaced by `6.2212554e-4` under the frozen convention.
5. CF-depth is not the hyperbolic depth `2 log q`. Both are now measured, but a mean shift is not an asymptotic metric-localization theorem.

6. The volume and counting-measure cut problems are distinct.
7. Version 3 preserves the exact public Version 2 parent, resolves localization data provenance, adds the bounded same-system multilevel diagnostic, and supplies a searchable publication PDF. A broader weighted-Farey prior-art scan and independent mathematical review remain valuable future work, but they do not block this bounded final-preprint successor.
8. No result here proves RH, changes Paper 3's spectrum-blind arithmetic deficit, proves full SIT, or authorizes a public action.

Appendix A. Correction provenance and cross-paper contract

The lower-bound source must be cited as BC-045-c24-audit-and-sharp-gap-lower-bound-20260612.md, not as bare `BC-045` and not through BC-045-claim-signature-20260612.*.

When separately authorized correction packages are prepared for Papers 1, 3, and 9, the required sentence is:

> For the weighted Farey Laplacian, the reviewed C-24 repair proves `lambda_2(L_N) >= c/(N^4 log N)` for some unspecified absolute `c > 0`; no numerical lower-bound constant and no pure `N^-4` lower bound are proved.

Propagation targets remain unapplied and separately approval-gated:

Target	Current target artifact	Owner	State	Approval
Paper 1, DOI 10.5281/zenodo.21231873	v5ma.github.io/production-canon/papers/branch-c-series/drafts/paper-01-farey-laplacian/v6-final.md	Micah Blumberg	`not-applied`	author approval required
Paper 3, DOI 10.5281/zenodo.21231877	v5ma.github.io/production-canon/papers/branch-c-series/drafts/paper-03-saturation-law/v6-final.md	Micah Blumberg	`not-applied`	author approval required; preserve spectrum-blind deficit boundary
Paper 9, DOI 10.5281/zenodo.21231894	v5ma.github.io/production-canon/papers/branch-c-series/drafts/paper-09-survey/v6-final.md	Micah Blumberg	`not-applied`	author approval required

Paper 3 must additionally state that the bracket bounds `||L_N^+||` but does not prove, alter, or strengthen its spectrum-blind RH-equivalent arithmetic deficit. Paper 8 may host a replay bundle but cannot promote a finite measurement to a theorem. This Version 3 package does not edit any companion paper or authorize any provider or mirror action.

Propagation to companion papers is explicitly deferred. This paper is self-contained, and no statement here depends on those future corrections.

The governed replay at reproducibility/localization-n80/ binds its source, parameters, environment specifications, generated JSON / CSV data, and runtime record through SHA-256. The separated reproducibility/localization-multilevel/ bundle applies the same system at four levels and discloses the feasibility probe. Together they close the local single-level and same-system multilevel reproduction gates only. They do not constitute independent replication, a preregistered trend test, a sharp asymptotic theorem, or authorization for a repository release.

Approval and release state

This manuscript is the locally final Version 3 preprint successor to public Version 2, Zenodo record 21330770, under concept DOI 10.5281/zenodo.21231880. Its theorem boundary, historical BC-010 disposition, executable replay, metadata, and PDF are frozen together in the Version 3 release package. External review remains invited. No Zenodo, OSF, mirror, or other repository release is performed by this manuscript package.

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Machine-readable bibliography expanded for this release

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